

Structured Training Program: Wi-Fi

Module 3: Introduction and History of Wi-Fi

1. What are the different 802.11 PHY layer standards? Compare their characteristics.

The IEEE 802.11 family defines several PHY (Physical Layer) standards, each evolving for higher data rates, better efficiency, and new features. Key ones include:

- 802.11 (Legacy, 1997): 2.4 GHz, DSSS/FHSS, max 2 Mbps, 20 MHz channel.
- 802.11b (1999): 2.4 GHz, HR-DSSS (with CCK), max 11 Mbps, 20 MHz. Good range but low speed, prone to interference.
- 802.11a (1999): 5 GHz, OFDM, max 54 Mbps, 20 MHz. Better speed, less interference than 2.4 GHz, but shorter range.
- 802.11g (2003): 2.4 GHz, ERP-OFDM (backward compatible with DSSS), max 54 Mbps, 20 MHz. Combines a-like speed with b-like range.
- 802.11n (Wi-Fi 4, 2009): 2.4/5 GHz, OFDM + MIMO (up to 4 streams), channel bonding (20/40 MHz), max ~600 Mbps. Introduced spatial multiplexing for higher throughput.
- 802.11ac (Wi-Fi 5, 2013): Mainly 5 GHz, OFDM, wider channels (up to 160 MHz), 256-QAM, MU-MIMO (downlink), max multi-Gbps (e.g., ~3.5 Gbps theoretical with 8x8).
- 802.11ax (Wi-Fi 6, 2019): 2.4/5 GHz (6 GHz in Wi-Fi 6E), OFDMA, 1024-QAM, MU-MIMO (UL/DL), longer symbols, flexible GI, max ~9.6 Gbps. Focus on efficiency in dense environments.
- 802.11be (Wi-Fi 7, 2024): 2.4/5/6 GHz, 320 MHz channels, 4096-QAM, enhanced MU-MIMO/OFDMA, multi-link operation, max theoretical ~46 Gbps.

Comparison of key characteristics:

1. Frequency Bands:
 - a. Early (b/g) - mainly 2.4 GHz (crowded, good range/penetration).
 - b. Later (a/ac) - 5 GHz (less interference, shorter range). Modern - adds 6 GHz (cleaner, widest channels).
2. Modulation & Access:
 - a. DSSS/FHSS (early)
 - b. OFDM (from a/g onward)
 - c. OFDMA (ax+).
3. Channel Width:
 - a. 20 MHz (early)
 - b. up to 40/80/160/320 MHz (bonding for speed).
4. MIMO:
 - a. None (early)
 - b. SU-MIMO (n)
 - c. MU-MIMO (ac/ax/be).
5. Max Data Rate: Increases dramatically with wider channels, higher QAM, more streams, and efficiency features.

6. Strengths/Trade-offs:

- a. 2.4 GHz = range + interference
- b. 5/6 GHz = speed + capacity.

Newer standards prioritize efficiency (density, latency) over raw peak speed in real-world use.

2. What are DSSS and FHSS? How do they work?

DSSS (Direct Sequence Spread Spectrum) and FHSS (Frequency Hopping Spread Spectrum) are spread-spectrum techniques used in early 802.11 to combat interference and improve reliability.

FHSS: The transmitter rapidly switches ("hops") the carrier frequency across a wide band according to a pseudo-random pattern (known to receiver). Data is sent in short bursts on different frequencies.

How it works: Reduces narrowband interference impact (only one hop affected) and provides some security (hard to jam without knowing pattern). Used in original 802.11. Lower data rates, better for certain mobility/low-interference scenarios.

DSSS: The data signal is multiplied (XOR) with a high-rate pseudo-random noise (PN) code (chipping sequence), spreading the signal over a wider bandwidth (e.g., 22 MHz channel). Receiver "despreads" using the same code to recover data.

How it works: Provides processing gain against interference/noise (interferers get spread out and appear as noise). Supports higher rates than basic FHSS. Used in 802.11/802.11b (with CCK for 5.5/11 Mbps).

In simple terms,

FHSS hops in time/frequency (good anti-jamming, lower throughput); DSSS spreads continuously across frequencies (higher rates, robust to narrowband noise, but more susceptible to wideband interference in theory). DSSS became dominant in WiFi for higher speeds.

3. How do modulation schemes work in the PHY layer? Compare different modulation schemes and their performance across various Wi-Fi standards.

Modulation encodes bits onto the carrier signal by varying amplitude, phase, and/or frequency.

Common schemes in WiFi:

- BPSK (Binary PSK): 1 bit/symbol (phase: 0° or 180°). Most robust (largest distance between points), lowest rate. Used for control/preamble or poor SNR.
- QPSK (Quadrature PSK): 2 bits/symbol (4 phases). Good balance of rate and robustness.
- QAM (Quadrature Amplitude Modulation): Combines amplitude + phase. 16-QAM (4 bits), 64-QAM (6 bits), 256-QAM (8 bits), 1024-QAM (10 bits), 4096-QAM (12 bits). Higher order = more bits/symbol means higher data rate, but constellation points closer means more susceptible to noise (requires higher SNR).

Performance across standard ds:

- Early (a/g): BPSK/QPSK/16-QAM/64-QAM with OFDM. Rates 6–54 Mbps. Lower orders for robustness.
- n/ac: Same + higher coding rates + MIMO. 256-QAM in ac for higher throughput.
- ax/be: Up to 1024/4096-QAM. Adaptive modulation (link adaptation chooses based on SNR). Higher orders need excellent channel conditions.

Trade-offs: Higher-order modulation (e.g., 256-QAM vs BPSK) increases spectral efficiency and data rate dramatically but raises Bit Error Rate (BER) in noisy/fading channels. Performance improves with better SNR, MIMO diversity, and error correction. In practice, WiFi uses fallback to robust schemes in poor conditions.

4. What is the significance of OFDM in WLAN? How does it improve performance?

OFDM (Orthogonal Frequency Division Multiplexing) divides a wide channel into many closely spaced orthogonal subcarriers (e.g., 52 in 802.11a, with 48 data). Each subcarrier carries a low-rate data stream using simple modulation (BPSK/QAM). Subcarriers overlap but don't interfere due to orthogonality.

Significance & Improvements:

- High spectral efficiency as Overlapping subcarriers pack more data.
- Resilience to multipath/fading because each subcarrier experiences flat fading (narrowband); cyclic prefix (guard interval) absorbs delay spread, reducing Inter-Symbol Interference (ISI).
- Robust to interference as narrowband interference affects few subcarriers.
- Equalization is simple. One-tap per subcarrier vs complex equalizers in single-carrier systems.
- Enabled jump from ~11 Mbps (b) to 54 Mbps (a/g) and paved way for wider channels/MIMO in later standards.

OFDM is the foundation for all modern WiFi PHY (from 802.11a onward). It dramatically improves performance in indoor multipath environments typical of WLANs.

5. How are frequency bands divided for Wi-Fi? Explain different bands and their channels.

WiFi uses unlicensed ISM/U-NII bands, such as

1. 2.4 GHz: 2.400–2.4835 GHz. 14 channels (5 MHz spacing, 20 MHz wide; non-overlapping: 1,6,11). Overlaps cause interference. Good range/penetration, but crowded (microwaves, Bluetooth, etc.). Supports up to 40 MHz bonding.
2. 5 GHz: Multiple sub-bands (U-NII-1 to U-NII-3/4). Many 20 MHz channels; supports 40/80/160 MHz bonding. Less interference, higher speeds, shorter range. Some channels require DFS (radar avoidance).
3. 6 GHz (Wi-Fi 6E/7): Large contiguous spectrum (~1200 MHz). Supports widest channels (up to 320 MHz). Cleanest band, highest capacity/speeds, but shortest range/highest attenuation.

Channels numbered by center frequency. Wider bonding (e.g., 80 MHz = 4×20 MHz) increases throughput but reduces available non-overlapping channels. Generally 2.4 GHz for coverage/IoT and 5/6 GHz are used for performance/density.

6. What is the role of Guard Intervals in WLAN transmission? How does a short Guard Interval improve efficiency?

In OFDM, a Guard Interval (GI) or Cyclic Prefix (CP) is inserted between symbols. It is a copy of the end of the symbol prepended to the beginning. This absorbs multipath delay spread, preventing ISI from previous symbols overlapping into the next.

- Provides time for echoes to die out before the next symbol arrives. Standard (long) GI: 800 ns.
- Short GI (400 ns, introduced in 802.11n): Reduces overhead. In good channel conditions (low delay spread, e.g., open spaces, short distances), it allows ~11% higher data rate (more symbols per second) with minimal error increase. Efficiency gain comes from less "wasted" time per symbol.
- Caveat: In high-multipath environments (e.g., indoors with reflections), short GI can increase errors so we use long GI.

7. Describe the structure of an 802.11 PHY layer frame. What are its key components?

The PHY frame is called PPDU (PLCP Protocol Data Unit). It encapsulates the PSDU (MAC frame + FCS).

Key components (varies slightly by standard, but general structure):

- Preamble: Synchronization (STF - Short Training Field for timing/frequency sync; LTF - Long Training Field for channel estimation). Legacy-compatible fields for backward compatibility.
- PHY Header (SIG fields): Contains rate, length, modulation/coding info, bandwidth, etc. (e.g., L-SIG, HT-SIG, VHT-SIG, HE-SIG in newer standards). Allows receiver to decode the rest.
- Data Field (PSDU): The actual MAC payload (MPDU), plus service bits, tail bits, padding. Includes the user data + MAC header/FCS.
- Optional fields (newer): PE (Packet Extension), midambles, etc

8. What is the difference between OFDM and OFDMA?

OFDM (Orthogonal Frequency Division Multiplexing): Divides channel into subcarriers for one user at a time. The entire channel is allocated to a single transmission (SU). Used in 802.11a/g/n/ac.

OFDMA (Orthogonal Frequency Division Multiple Access): Extension of OFDM. Divides subcarriers (into Resource Units - RUs) and assigns different RUs to multiple users simultaneously in frequency and time. Supports uplink/downlink multi-user access.

9. What is the difference between MIMO and MU-MIMO?

MIMO (Multiple Input Multiple Output): Uses multiple antennas at Tx and Rx for spatial multiplexing (send multiple independent data streams simultaneously on same frequency). Improves throughput and/or reliability via diversity (e.g., 2x2, 4x4). Mainly Single-User (SU-MIMO) in 802.11n.

MU-MIMO (Multi-User MIMO): Extends MIMO to serve multiple users at once. AP uses beamforming to direct streams to different clients simultaneously (spatial separation). Used in Downlink in 802.11 ac; Used in UL+DL in 802.11 ax.

10. What are PPDU, PLCP, and PMD in the PHY layer?

PPDU (PLCP Protocol Data Unit): The complete PHY frame transmitted over the air (Preamble + Header + PSDU).

PLCP (Physical Layer Convergence Procedure): Sub-layer that bridges MAC and PMD. Takes PSDU from MAC, adds preamble/header, and prepares it for transmission (convergence function).

PMD (Physical Medium Dependent): The actual transceiver sub-layer. Handles modulation, demodulation, RF transmission/reception, and medium-specific aspects (e.g., DSSS vs OFDM details).

PLCP is "format agnostic" to the medium; PMD is hardware-specific. Together with PLCP, they form the PHY.

11. What are the types of PPDU? Explain the PPDU frame format across different Wi-Fi generations.

Types of PPDU:

1. Legacy(Non-HT): DSSS (long/short preamble), OFDM (802.11a/g).
2. HT (High Throughput, 802.11n): Mixed/ Greenfield formats with HT-SIG.
3. VHT (Very High Throughput, 802.11ac): VHT preamble for wider channels/MU-MIMO.
4. HE (High Efficiency, 802.11ax): HE PPDU with U-SIG, HE-SIG, support for OFDMA, longer symbols.
5. EHT (Extremely High Throughput, 802.11be): Further enhancements with multi-link, 320 MHz, etc.

General Format Evolution:

- Legacy: Simple preamble (STF/LTF) + SIG + Data.
- Modern: Legacy-compatible portion (L-STF, L-LTF, L-SIG) for coexistence + repeated SIG + high-efficiency fields (e.g., RL-SIG, U-SIG, user-specific SIG) + Data + PE.
- This ensures backward compatibility while adding features. Preamble length and complexity increase with capabilities.

12. How data rate is calculated ?

Data rate (PHY rate) depends on:

Symbol rate = $1 / (\text{Symbol Duration})$. Symbol Duration = Useful symbol time + Guard Interval.

Bits per symbol = Modulation order (e.g., 6 for 64-QAM) $\times$ Coding rate (e.g., 3/4) $\times$ Number of data subcarriers.

Spatial streams (MIMO multiplier).

Channel width (more subcarriers in wider channels).

Formula (OFDM):

Data Rate = (Number of data subcarriers × Bits per subcarrier × Coding rate × Spatial streams) / Symbol Duration

Example (802.11a, 20 MHz, 64-QAM 3/4, long GI): ~54 Mbps.

Short GI or wider channels/MIMO increase it (e.g., ~11% from short GI). MCS (Modulation and Coding Scheme) index tables in standards summarize rates for combinations.